

Anti-UAV410: A Thermal Infrared Benchmark and Customized Scheme for Tracking Drones in the Wild

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Abstract—The perception of drones, also known as Unmanned Aerial Vehicles (UAVs), particularly in infrared videos, is crucial for effective anti-UAV tasks. However, existing datasets for UAV tracking have limitations in terms of target size and attribute distribution characteristics, which do not fully represent complex realistic scenes. To address this issue, we introduce a generalized infrared UAV tracking benchmark called *Anti-UAV410*. The benchmark comprises a total of 410 videos with over 438 K manually annotated bounding boxes. To tackle the challenges of UAV tracking in complex environments, we propose a novel method called Siamese drone tracker (*SiamDT*). *SiamDT* incorporates a dual-semantic feature extraction mechanism that explicitly models targets in dynamic background clutter, enabling effective tracking of small UAVs. The *SiamDT* method consists of three key steps: *Dual-Semantic RPN Proposals (DS-RPN)*, *Versatile R-CNN (VR-CNN)*, and *Background Distractors Suppression*. These steps are responsible for generating candidate proposals, refining prediction scores based on dual-semantic features, and enhancing

the discriminative capacity of the trackers against dynamic background clutter, respectively. Extensive experiments conducted on the *Anti-UAV410* dataset and three other large-scale benchmarks demonstrate the superior performance of the proposed *SiamDT* method compared to recent state-of-the-art trackers.

Index Terms—Anti-UAV, siamese network, single object tracking, thermal infrared tracking dataset, tiny target tracking.

I. INTRODUCTION

WITH the rapid advancements in deep learning and automation, Unmanned Aerial Vehicles (UAVs) have become ubiquitous and widely used in various fields such as aerial photography and environmental monitoring [1]. However, along with their widespread adoption, UAVs also pose potential threats to public safety. Therefore, it is crucial to monitor their operations, including their locations and trajectories, in order to ensure safety and security. However, due to the small size of UAVs and the complexity of their surrounding environments, locating and tracking UAVs remains a challenging problem that requires careful consideration.

Traditional datasets for thermal infrared object tracking [2], [3], [4], [5], [6] primarily emphasize tracking of large-scale targets, which may not be ideal for small object tracking scenarios such as UAVs. Although the *Anti-UAV* dataset [2] has been introduced to address UAV tracking, it still exhibits limitations in terms of realistic scene representations. The drones in this dataset are often susceptible to easy observation due to their large size and clean backgrounds, which do not fully capture the complexity of real-world tracking situations.

In order to address this limitation, we have developed a new dataset called *Anti-UAV410*, which is currently the largest benchmark for single object tracking of UAVs in the thermal infrared (TIR) domain. This dataset comprises 410 sequences with over 438 k bounding boxes in total, and is specifically designed to tackle UAV tracking challenges. *Anti-UAV410* focuses on restoring the anti-UAV tracking problem in real scenarios, i.e., small UAV object tracking under long-range imaging. As shown in Fig. 1 (left), *Anti-UAV410* covers various tracking scenes in the wild, such as forests, mountains, lakes, which may introduce excessive background noises and make the tracker learn from the background region. Thereby *Anti-UAV410* gives rise to new challenges in tracking UAV in the wild, i.e. tiny targets and dynamic backgrounds with clutter (DBC). As depicted in Fig. 1 (right), the dataset includes 77 video sequences associated with the DBC challenge and more than 100 video sequences containing tiny-sized drones, which provides better data support for

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The benchmark of *Anti-UAV410* is available at <https://github.com/HwangBo94/Anti-UAV410>.

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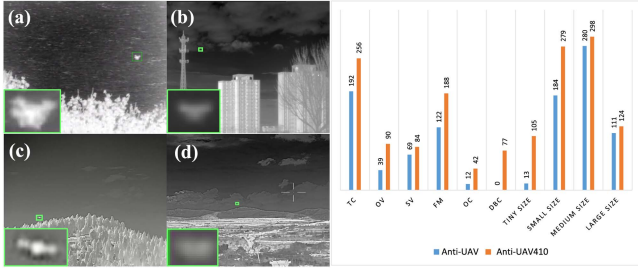


Fig. 1. Example frames from the Anti-UAV410 dataset, depicting realistic backgrounds including lakes in (a), buildings in (b), forests in (c) and hills in (d). The bar chart on the right compares the attributes of the Anti-UAV [2] and Anti-UAV410 datasets, highlighting their relevance for practical anti-UAV tasks in terms of video count.

anti-drone missions and serves as a comprehensive platform for both training and evaluating tracking algorithms.

As a solution to the challenges posed by tiny object tracking against dynamic and ambiguous backgrounds, we propose a customized multi-branched tracking architecture called *SiamDT* (Siamese Drone Tracker). Due to the missing color features in thermal infrared (TIR) images, conventional TIR tracking methods often utilize shallow features such as contours for target representation [7], [8], [9], [10], [11], which are highly interpretable. However, given that UAVs are typically small in size and operate in cluttered environments, these shallow hand-crafted features are insufficient for accurately distinguishing UAVs from background distractors with similar visual appearances.

To bridge these gaps, we design a novel dual-semantic extracting mechanism to capture semantic saliency of the target that are discriminative and easily localized in the search template. The dual-semantic extracting mechanism consists of two-stages modules: i) *Dual-Semantic RPN Proposals (DS-RPN)* that predicts candidate proposals by constructing a Siamese-branched RPN that models relations between the target region and template as well as the probability of the existence of foreground object; ii) *Versatile R-CNN (VR-CNN)* that refines the prediction proposals based on the fused information through shared weights' R-CNNs. Different from traditional Siamese trackers [12], [13], [14] that only learn the matching semantic between template and target candidate regions during training, *SiamDT* also learns an individual foreground probability semantic to determine the probability of the current target candidate region containing a foreground object. Specifically, we introduce an additional branch to output the probability of foreground semantic in the current target region, which is nearly zero when the target candidate regions have no semantic object. Based on the robust feature representation, we further introduce the third branch iii) *Background Distractors Suppression* that enhances trackers' discriminative capacity against dynamic background distractors by a hand-crafted map that stores all possible negative samples. Through joint action of step 1 and step 2, an inductive representation with explicit semantics about the small targets can be obtained in a superior way. Consequently, our *SiamDT* copes with two main challenges during tracking UAVs

in realistic-like scenes, undersize target problem and blurred background problem.

Extensive experimentation was conducted to analyze the performance of state-of-the-art trackers in addressing various challenges. In summary, this work contributes in the following ways:

- We introduce a large-scale Thermal Infrared (TIR) benchmark called Anti-UAV410, which serves as a valuable resource for anti-UAV systems and tracking methods. Furthermore, we assess the performance of existing state-of-the-art trackers on Anti-UAV410 and provide comprehensive baselines for future comparisons.
- We propose a novel Siamese drone tracker, named *SiamDT*, which leverages powerful dual-semantic features to enhance discrimination of tiny targets and includes a background suppression branch to mitigate dynamic background clutter. *SiamDT* presents a novel scheme for TIR tracking of small targets.
- We achieve new state-of-the-art performance in tracking on the newly proposed Anti-UAV410 benchmark as well as the Anti-UAV [2] dataset, and demonstrate the generalization ability on other generic datasets such as LaSOT [15] and GOT-10k [16].

II. RELATED WORK

A. Anti-UAV Tracking

UAV tracking tasks can be divided into two categories: one is aerial viewpoint tracking and the other is anti-UAV tracking. Although they are both associated with drones, there is a fundamental difference between these two tracking tasks.

The aerial viewpoint tracking employs UAVs as the collecting devices and finds the region of interest from an air-to-ground perspective, with no limitations on the tracking targets. The aerial tracking task has gained a great deal of attention and achievements in recent years. Dronescape [17] presents a large-scale UAV video dataset from real scenarios for multi-task learning. UAV123 [18] and UAVDT [19] construct UAV benchmarks focusing on tracking objects in videos captured by a UAV platform. UAVDark135 [20], DarkTrack2021 [21] and NAT2021 [22] are the pioneer datasets for nighttime aerial tracking, and an associated Transformer based UAV nighttime tracker is proposed in [23].

The anti-UAV tracking, on the other hand, fixes the tracked target as a drone to simulate a real anti-UAV scenario, aiming at solving the realistic problem of tracking drones. Therefore, aerial viewpoint tracking and anti-UAV tracking are essentially two distinct tracking tasks, which are applicable in different tracking scenarios. Jiang et al. [2] proposed a multi-modal anti-UAV dataset, consisting of 318 pairs of TIR-visible video sequences, providing valuable resources for anti-UAV tracking research. Nevertheless, the targets in the dataset are mostly large-sized, and the tracking scenes mainly comprise skies or urban buildings, which may not be as complex as real-world scenarios. Zhao et al. [24] proposed the DUT dataset, which contains 10,000 visible UAV images for the detection task and 20 visible sequences for the tracking task. Unfortunately, the

amount of tracking portion of the DUT dataset is far less to be used for training and evaluation of anti-UAV algorithms. To overcome the limitations method above, we propose the Anti-UAV410 dataset, which includes more realistic scenes to better simulate actual tracking scenarios.

B. Thermal Infrared Datasets

Early thermal infrared (TIR) datasets have primarily been used for pedestrian detection [25], [26]. In recent years, with the advancements in TIR sensors, several TIR datasets have been proposed, which have contributed to the development of object tracking. Berg et al. [27] introduced the first TIR benchmark (VOT-TIR2015) for tracking, consisting of 20 thermal image sequences with different objects, collected from seven different sources using eight different types of sensors. VOT-TIR2016 [5] extends VOT-TIR2015 by opening a call for contributions and replacing the simple sequences with community-provided sequences. VOT-RGBT2019 [4] is a dual-mode datasets, containing total 234 sequences with an average length of 335 frames, which aims to make full use of RGB and thermal image information to enhance the tracker's performance. Liu et al. [6] proposed a large-scale dataset (LSOTB-TIR) with a wider range of objects and scenarios, posing greater challenges for tracking. LSOTB-TIR consists of 1,400 TIR sequences with more than 600 K frames, which is the largest TIR object tracking dataset to date. Zhang et al. [28] proposed using Generative Adversarial Networks (GANs) to generate synthetic TIR data from abundant labeled RGB data, addressing the issue of limited TIR training samples.

However, there is a scarcity of datasets specifically designed for UAV tracking, which hinders the development of anti-UAV tasks. Recently, Jiang et al. [2] proposed a multi-modal UAV dataset, but both of the target size and the target's background are deviating from the actual tracking scenario. To bridge the above gap, we propose the Anti-UAV410 dataset, offering a platform for training and evaluation of anti-UAV tracking algorithms.

C. Thermal Infrared Trackers

Feature extraction, which captures fine-grained semantic information, plays a crucial role in object perception. Several methods have been proposed in the literature for feature extraction in TIR tracking. Yu et al. [29] modified the Histogram of Oriented Gradient (HOG) features based on the profile information of TIR targets and trained the tracker with dense samples, achieving promising performance in TIR tracking. Zhang et al. [7] trained a Siamese network to extract TIR features and demonstrated the effectiveness of shallow features of infrared targets in differentiating trackers. Huang et al. [8] introduced a spatio-temporal attention constraint to suppress the interference of cluttered backgrounds, thereby enhancing the tracker's ability to cope with tracking texture-less infrared targets in cluttered backgrounds. Chen et al. [30], [31] extracted handcrafted features such as motion and edge features to enhance the discriminative ability of TIR trackers. Burceanu et al. [32] proposed a dual-pathway tracking algorithm, which used one

pathway to resolve noise interference and the other pathway to accommodate changes in target appearance, which provides a good idea for tracking infrared UAV targets with severe noise interference. Zhang et al. [33] provides a small target dataset (TSFMO) as well as a dedicated small target tracker (S-KeepTrack). Although S-keepTrack is designed for RGB video sequences, it can handle the challenges of small targets and fast movements, and thus it has application potential for tracking anti-drone in infrared images. Jiang et al. [2] proposed a novel training strategy that helps the tracker learn more robust class-level semantic information and obtain more discriminative instance-level features for TIR tracking. However, most of these methods primarily focus on large-sized TIR targets, and little attention has been paid to tiny target tracking. To address this issue, we develop an efficient feature extraction method and a background distractors suppression strategy for TIR tiny target tracking. These techniques aim to capture fine-grained semantic information and suppress the interference from cluttered backgrounds, enabling more accurate and robust tracking of small-sized targets in TIR images.

III. ANTI-UAV410 BENCHMARK

Traditional datasets typically prioritize tracking of large-sized objects against clean backgrounds, which may not fully capture the real-world scenarios of drones that often operate in cluttered backgrounds as small objects. To address this limitation, we propose Anti-UAV410, a dataset that is designed to be more representative of realistic scenes, featuring numerous tracking scenes with tiny targets and dynamic backgrounds.

A. Data Collection

To create a realistic representation of the challenges of anti-UAV tracking in real-world environments, we captured tracking videos in diverse and complex scenarios. These scenarios included two different lighting conditions (day and night), two seasons (autumn and winter), and a wide range of backgrounds such as buildings (30%), mountains (20%), forests (5%), urban areas (30%), clouds (10%), water surfaces (3%), and others. The acquired video sequences were recorded at a frame rate of 25 frames per second (FPS) in the mid-wave infrared spectrum. From these sequences, we selected 100 minutes of video for meticulous frame-by-frame annotation, totaling over 150,000 frames.

To further augment the size of the dataset, we also incorporated videos from the 1st Anti-UAV Challenge website [34] and the Anti-UAV dataset [2]. These videos were cropped and cleaned to ensure consistency and quality, resulting in the creation of the Anti-UAV410 dataset. i) Since some video sequences have text on their borders, we adopted the crop operation to remove these distractions. ii) Because Anti-UAV is a dual-modal dataset, some of the video fragments are not suitable for individual TIR tracking (e.g., some of the TIR video has no target in the first frame, while the corresponding visible video does). We discard such frames so that it can be used for independent TIR tracking. We call this process the "clean" operation. The average length of the videos in the Anti-UAV410 dataset is 1069

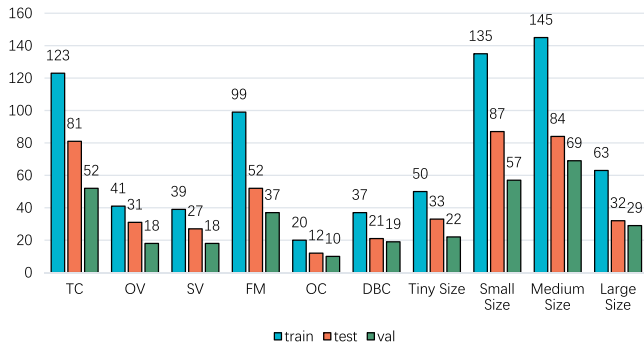


Fig. 2. Statistical distribution of the training set, test set and validation set under each attribute. The vertical coordinate indicates the number of videos, and the three subsets are relatively evenly distributed.

frames, which poses a challenge for trackers to have robust target re-detection capabilities.

B. Annotation

Following the annotation process described in [2], we initially manually annotated the attributes of each sequence, including the size of UAVs and the presence of any UAVs. Subsequently, we annotated the UAV targets using external rectangle bounding boxes as the ground truth. The coordinates of the top-left corner, width, and height of the bounding box were recorded to represent the ground truth of the target. In cases where there was no target present in a frame, the ground truth was represented as empty.

C. Statistics

Dataset Splitting: The Anti-UAV410 dataset is divided into three sets: the train set, which consists of 200 sequences; the validation set, which consists of 90 sequences; and the test set, which consists of 120 sequences. Unlike previous datasets, we have carefully divided the dataset into three sets based on the attributes of the sequences. The attribute distributions of the train, test and validation sets, are as shown in the Fig. 2, the three subsets are relatively evenly proportioned on each challenge attribute, and at this point using the training set can better learn the challenges faced by drone tracking in the wild scenarios. No special technical, we make the Anti-UAV410 dataset exhibit such distributions by manually adjusting.

Furthermore, the training set and validation set are derived from non-overlapping clips of the same sequence, while the test set is entirely independent, ensuring a rigorous evaluation of tracker performance.

Scale distribution: In order to replicate the challenges of anti-UAV tracking in realistic scenarios, we have focused on increasing the proportion of small targets in our Anti-UAV410 dataset. The frame size of all sequences in Anti-UAV410 is set at 640×512 , and we define the target size as the diagonal length of the bounding boxes. To compare the target size distribution with other widely used TIR datasets such as PTB-TIR [26], LSOTB-TIR [6], VOT-TIR2015 [35], VOT-TIR2016 [5], Anti-UAV [2], and VOT-RGBTIR2019 [4], we have analyzed the distribution in terms of target size. Unlike other tracking tasks, there are

TABLE I
ILLUSTRATION OF ATTRIBUTES IN ANTI-UAV410

Attribute	Description
OV	Out-of-View (90): the target leaves the view.
OC	Occlusion (42): the target is partially or heavily occluded
TC	Thermal Crossover (256): the target has a similar temperature to other objects or background surroundings.
FM	Fast Motion (188): the target's motion between two adjacent frames is larger than 60 pixels.
SV	Scale Variation (84): the ratio of the bounding boxes of the first frame and the current frame is out of the range [0.66,1.5].
DBC	Dynamic Background Clutter (77): there are dynamic changes in the background around the target

The number in brackets represents the number of sequences for the corresponding attribute.

TABLE II
COMPARISON OF SOME POPULAR TIR DATASETS AND ANTI-UAV410, THE TARGET SIZE IS CALCULATED BY THE DIAGONAL OF THE BOUNDING BOXES, AND THE FRAME NUMBERS OF CORRESPONDING SIZE ARE REPORTED

Dataset	Tiny	Small	Normal	Large
LSOTB-TIR	0	7625	30910	568294
VOT-TIR2015	0	480	2165	8624
VOT-TIR2016	0	1681	3037	9145
VOT-RGBTIR2019	283	790	4849	14161
PTB-TIR	0	1883	6325	73925
Anti-UAV	858	6419	88181	198317
Anti-UAV410	17071	126649	97942	196735

rarely UAV targets with scales larger than 70×70 pixels in anti-UAV tracking missions. Given the inherently small scale of drones, we define the tiny-size UAV target more strictly. We first set the size divider for tiny and small targets to 10 pixels. To distribute the different scale attributes as evenly as possible, we then define the size range into 4 intervals: tiny [2, 10), small [10, 30), medium [30, 50) and normal [50, ∞). The comparison results are presented in Table II. Notably, the Anti-UAV410 dataset exhibits a significant proportion of small targets, with more than half of the targets having sizes less than 50 pixels, and even includes a certain percentage of tiny targets (size less than 10 pixels), which are missed by other datasets such as VOT-TIR2015 [35], VOT-TIR2016 [5], PTB-TIR [26], and LSOTB [6].

Attribute Definition: To comprehensively evaluate the performance of the trackers, we have labeled each sequence in our dataset with six attributes, namely Out-of-View (OV), Occlusion (OC), Thermal Crossover (TC), Fast Motion (FM), Scale Variation (SV), and Dynamic Background Clutter (DBC), as shown in Table I. The Anti-UAV410 dataset contains a significant number of thermal crossover situations, which are commonly encountered in TIR sequences. Additionally, due to the motion characteristics of UAVs, our dataset includes considerable fast movements, making the tracking task more challenging. Furthermore, Anti-UAV410 covers a substantial amount of dynamic background clutter, which is often present in real-world scenarios. It is noteworthy that the target sizes in Anti-UAV410 are predominantly tiny, small, and medium sizes, which are

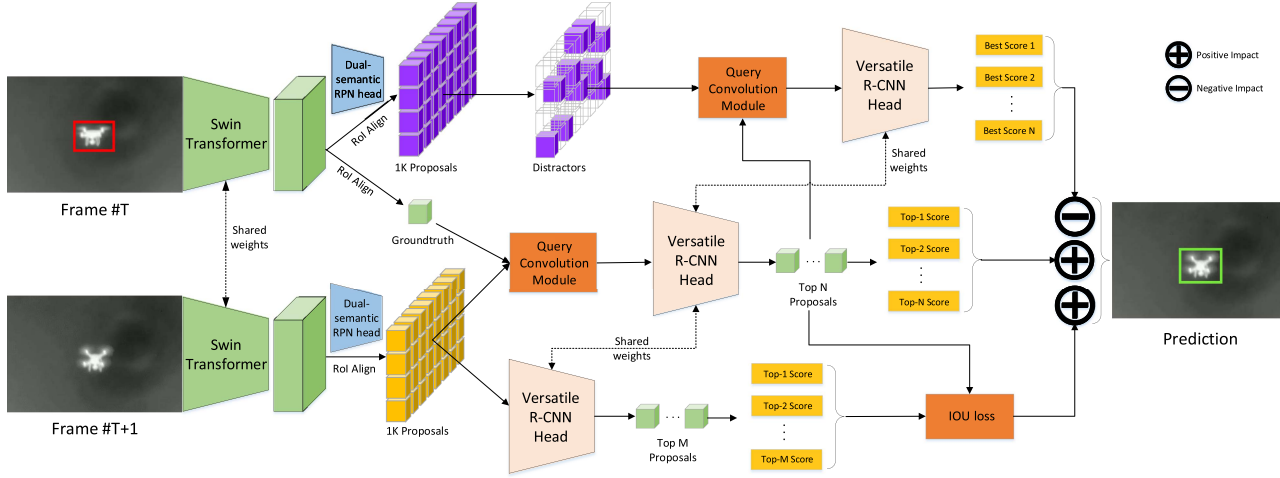


Fig. 3. Overview of SiamDT, which comprises of the *Swin Transformer backbone*, *Dual-semantic RPN network*, *Versatile R-CNN detection head*, and *Query Convolution Module*. Initially, the features of the template branch and the search branch are obtained through a Siamese feature extraction network. Subsequently, the features of both branches are separately inputted into the DS-RPN to generate the most promising candidate proposals. Finally, the optimal decision is made using the three branches of VR-CNN, namely the suppression of background distractors branch, the matching semantics branch, and the foreground probability semantics branch, in a top-to-bottom manner.

significantly smaller than those in other datasets, thereby posing a formidable challenge for trackers.

D. Compared With Anti-UAV Dataset

The Anti-UAV dataset [2] is a multi-modal dataset that includes 318 video pairs. This dataset is designed to address the challenge of anti-UAV tracking in different modalities, such as low-light conditions where vision-based trackers may perform poorly. However, our proposed Anti-UAV410 dataset is specifically tailored to address the long-range anti-UAV tracking problem. It aims to improve the tracker's robustness against background clutter and enhance its ability to discriminate small-sized UAV targets. Unlike the Anti-UAV dataset, which has relatively large target sizes (with an average diagonal length of 40 pixels) and homogeneous backgrounds (consisting mainly of sky and urban buildings), we have collected a substantial amount of wild anti-UAV tracking videos to replicate real-world scenarios with diverse backgrounds and challenging tracking conditions.

IV. SIAMDT TRACKER

A. Overview

Drawing inspiration from the success of Siamese trackers such as SiamATL [36], SiamFC [37], SiamRPN [13], DaSiamRPN [14], SiamBAN [38], and SiamCAR [39], we have developed our own Siamese drone tracker (SiamDT) using a multi-branched architecture, as depicted in Fig. 3. Specifically, SiamDT employs a Siamese feature extraction subnetwork that incorporates attention mechanisms and adopts the Swin Transformer [40] as the backbone feature extractor.

Distinguished from traditional Siamese trackers, the decision-making subnetwork of SiamDT consists of three specialized branches. One branch is designed to suppress background noise,

while another branch predicts the prior probability of the existence of semantic information targets in the target area. The last branch performs similarity evaluation between the detection image and the template. Importantly, these branches are supported by three specially designed modules: the Dual-Semantic RPN Proposals (DS-RPN), the Versatile R-CNN (VR-CNN), and the Query Convolution Module (which encodes the template to the candidate regions and predicts similarity scores through combining VR-CNN).

In the following sections, we will provide a detailed introduction to each module of the SiamDT architecture.

B. Dual-Semantic RPN for Proposals Generation

Most one-stage Siamese trackers [13], [14], [41] rely on vanilla RPN networks to establish associations between the searching image and the template. Although these vanilla RPN networks perform well in most cases, they tend to fail or make inappropriate tracking decisions when the search image is significantly larger than the template image due to their inherent regional searching strategy. In order to overcome this limitation, we propose a novel Dual-semantic RPN (DS-RPN) network with an additional branch that performs global searching. Experimental results demonstrate that our method is capable of accurately predicting optimal candidate proposals.

Global RPN Proposals: Inspired by the approaches used in GlobalTrack [42] and Siam R-CNN [12], we incorporate an RoI pooling layer to enable global RPN generation. Specifically, in the template branch, we extract the features of the target region and process them through an RoI alignment layer. In the detection branch, the global features are extracted from the entire test image, and an RPN network is used to generate a large number of candidate proposals. Each proposal is then aligned to the same size as the target features, allowing for global matching with the template target.

Dual-semantic Feature Extraction: Let z and x represent image patches of the template and searching area, respectively. The DS-RPN network incorporates two semantic information sources. The first is the foreground probability semantics of the target region, denoted as $\varphi(x) \in \mathbb{R}^{h \times w \times c}$, which indicates whether a target containing semantic information is present in the target region. The second is the matching semantics between the template and the target region. Denoting the features of the template region as $\varphi(z) \in \mathbb{R}^{k \times k \times c}$, the matching semantics $\varphi_{corr}(x, z)$ can be calculated as:

$$\varphi_{corr}(x, z) = (\text{Conv}_{\text{out}}(\text{Conv}_x(\varphi(x)) \otimes \text{Conv}_z(\varphi(z))) + b), \quad (1)$$

where $\otimes$ denotes the depth-wise convolution operator [41] used to compute the similarity between $\varphi(z)$ and $\varphi(x)$. Conv_z applies a $k \times k$ convolution layer with zero padding on $\varphi(z)$ to convert the features into a 1×1 convolutional kernel. Conv_x indicates the application of a 3×3 convolution layer with 1-pixel padding to the feature $\varphi(x)$. Conv_{out} denotes a 1×1 convolution layer that converts the channel number back to c . b represents the bias term.

Dual-semantic Proposals: Based on the foreground probability semantics and matching semantics as defined in (1), they are concatenated and input into the DS-RPN network to generate the candidate proposals DS_{prs} using the RPN head as follows:

$$DS_{prs} = \text{RPNHead}(\text{Concat}(\varphi_{corr}(x, z), \varphi(x))). \quad (2)$$

where Concat denotes the concatenation operation, and RPNHead represents the RPN head module that takes the concatenated features as input and generates the candidate proposals DS_{prs} for further processing in the tracker.

Finally, an ROI pooling layer [42] is exploited to obtain aligned global RPN proposals DS_{alg} :

$$DS_{alg} = \text{RoIPooling}(DS_{prs}). \quad (3)$$

where RoIPooling refers to the RoI pooling operation.

Dual-semantic RPN Loss Function: Similar to [43], our loss function for DS-RPN is defined as:

$$\begin{aligned} \mathcal{L}_{\text{DS-RPN}} &= \mathcal{L}_{\text{RPN}}(\text{Concat}(\varphi_{corr}(x, z), \varphi(x))) \\ &= \frac{1}{N_{cls}} \sum_i \mathcal{L}_{cls}(p_i, p_i^*) + \lambda \frac{1}{N_{reg}} \sum_i p_i^* \mathcal{L}_{reg}(b_i, b_i^*), \end{aligned} \quad (4)$$

where $p_i^* \in 0, 1$ and $p_i \in [0, 1]$ denote the ground truth label and predicted score of the i -th anchor, respectively. b_i^* and b_i represent the 4 parameterized coordinates of the ground truth and predicted bounding box, respectively. The two terms are normalized by N_{cls} and N_{reg} , and weighted by a balancing parameter λ .

C. Versatile R-CNN Module for Results Refine

Previous research [12], [42] has demonstrated that a two-stage network structure outperforms a single-stage one in tracking tasks. Motivated by this finding, we incorporate the second stage of R-CNN [44] and introduce a versatile R-CNN head

that is capable of effectively handling aligned proposals of dual-semantic features extracted in Section IV-B.

Versatile R-CNN Input: We aim to assess the similarity of candidate proposals to the template features while preserving the inherent semantic information of the drone targets. To achieve this, we first encode the target information into each proposal. Let $\Psi(x) \in \mathbb{R}^{n \times k \times k \times c}$ be the feature matrix of the RPN aligned proposals, where n denotes the number of proposals. We compute the new matching feature $\Psi_{corr}(x, z) \in \mathbb{R}^{n \times k \times k \times c}$ by encoding the template region feature $\Psi(z) \in \mathbb{R}^{1 \times k \times k \times c}$ as follows:

$$\Psi_{corr}(x, z) = \text{Conv}'_{\text{out}}(\text{Conv}'_x(\Psi(x)) \odot \text{Conv}'_z(\Psi(z))) + b, \quad (5)$$

where $\odot$ denotes the Hadamard product, which is utilized to fuse the information from both branches. Similarly, Conv'_x refers to performing a 3×3 convolution with 1 pixel padding on each channel of $\Psi(x)$. Conv'_z performs the same operation on $\Psi(z)$ as Conv'_x , except that Conv'_z automatically expands $\Psi(z)$ to n dimensions after the convolution. $\text{Conv}'_{\text{out}}$ is a 1×1 convolution layer with c output channels, and b represents the bias term.

Versatile R-CNN Head: Similar to the dual-semantic RPN, we can concatenate $\Psi_{corr}(x, z)$ and $\Psi(x)$ to create a dual-semantic R-CNN head, which exhibits strong discrimination. However, during the tracking phase, it is preferable to have independent predictions for foreground probability semantics and matching semantics for the target location. This approach mitigates the risk of a single semantic error resulting in complete tracking failure. Thus, we design a versatile R-CNN head that can output two scores. One score determines the presence of a target with semantic information in the target region, while the other represents the similarity between the template and the target region. These two processes operate independently, but share the same weights.

Versatile R-CNN Loss Function: During the training phase, we first input the original features $\Psi(x)$ into the R-CNN head and obtain a loss $\mathcal{L}_{\text{R-CNN}}(\Psi(x))$ for the classification of drones. Next, we feed the encoded features $\Psi_{corr}(x, z)$ (notably, $\Psi(x)$ and $\Psi_{corr}(x, z)$ have the same size and number of channels) into the R-CNN head, and acquire a matching loss $\mathcal{L}_{\text{R-CNN}}(\Psi_{corr}(x, z))$ with the template branch. The overall loss function of the Versatile R-CNN is then calculated as follows:

$$\begin{aligned} \mathcal{L}_{\text{VR-CNN}} &= \mathcal{L}_{\text{R-CNN}}(\Psi_{corr}(x, z)) + \mathcal{L}_{\text{R-CNN}}(\Psi(x)) \\ &= \frac{1}{N_{prop}} \sum_i \left\{ \mathcal{L}_{cls}(p_{i,corr}, p_i^*) + \mathcal{L}_{cls}(p_{i,orig}, p_i^*) \right. \\ &\quad \left. + \lambda p_i^* (\mathcal{L}_{reg}(b_{i,corr}, b_i^*) + \mathcal{L}_{reg}(b_{i,orig}, b_i^*)) \right\}, \end{aligned} \quad (6)$$

where N_{prop} is the total number of candidate proposals. $p_{i,corr}$ and $p_{i,orig}$ are the prediction confidence scores of matching semantics and foreground probability semantics, respectively. Similar to b_i in (4), $b_{i,corr}$ and $b_{i,orig}$ are the predicted bounding boxes of matching semantics and foreground probability semantics, respectively.

D. Three-Branch Network for Decision Making

The tracking process of our approach encompasses three branches that converge for final decision making:

Background Distractors Suppression Branch: In contrast to the conventional Siamese network framework that solely calculates the RPN proposals of the search branch, our approach performs RPN candidate proposal extraction for both branches. In the template branch, we identify distractors in the background region based on their high confidence scores. Specifically, we enforce a zero overlap rate between the bounding boxes of the distractors and the ground truth bounding boxes. Let D denote the set of background distractors, which consists of K proposals, where $D = \{d_1, d_2, \dots, d_K\}$. We limit the number of distractors to a maximum of 50, i.e., $K \leq 50$. In the search branch, multiple predictions may arise from the output of the R-CNN head due to the presence of background distractors, making it challenging to discriminate between them. Let P denote the set of N hard-to-discriminate candidate predictions, where $P = \{p_1, p_2, \dots, p_N\}$. We calculate the correlation score of the prediction results with the background distractors using the following formula:

$$G(D, P) = \sum_i R - \text{CNNHead}(\Psi_{\text{corr}}(D, p_i)), \quad (7)$$

where $G(D, P)$ computes the similarity between the prediction candidates P and the background distractors D . We use the maximum confidence score $G(D, p_i)$ of the R-CNN predictions as the similarity output of p_i to D . A higher value of $G(D, p_i)$ indicates a higher probability that p_i is a background distractor. Hence, we leverage $G(D, p_i)$ as a negative feedback to adjust the final prediction result.

The set D is initialized with the distractors from the first frame, and the features of each background distractor are stored in a sequence memory. If we obtain a reliable tracking result at a certain frame, we update D accordingly; otherwise, D remains unchanged.

Matching Semantic Branch: Following the approach of traditional Siamese trackers [12], [42], we perform a forward pass of DS-RPN, Query Convolution Module, and VR-CNN using the template image from the first frame and the search image of the current frame as input. Subsequently, we extract the top N proposals from the output.

Foreground Probability Semantic Branch: Considering the challenge of extracting effective feature representations for tiny targets with limited pixel occupancy, we introduce a third branch for foreground probability semantics. This branch is designed to output a prior probability value ranging from 0 to 1, indicating the likelihood of the current proposal containing the semantic UAV target. Subsequently, we select the top M proposals from this branch for the final decision-making process.

Remark 1: During the final decision stage, we obtain N proposals from the matching semantic branch, each with a matching semantic score, as well as similarity scores of background distractors from the background distractors suppression branch. Additionally, the foreground probability semantic branch outputs M proposals that may contain the semantic UAV target,

and calculates IoU scores between these proposals and the previous N proposals. These three scores are linearly combined, and the proposal with the highest score is selected as the final tracking result.

V. EXPERIMENTS

We conduct a qualitative and quantitative comparison between our proposed SiamDT tracker and 25 state-of-the-art trackers on our Anti-UAV410 dataset. We also analyze in detail the performance of each tracker under different attribute challenges, and discuss their potential contributions to the advancement of the anti-UAV tracking in the wild. Furthermore, we conduct experiments on other RGB datasets to validate the generalization ability of SiamDT in RGB tracking.

A. Experimental Settings

Implementation Details: For the Siamese feature extraction subnetwork, we utilize the Swin Transformer architecture [40] as the backbone. The hyperparameters including the size of the shifted window, query dimension of each head, and the expansion layer of each multi-layer perception (MLP) are set to 7, 32, and 4, respectively. SiamDT takes Swin-Tiny as the backbone by default, its hidden channels are set to 96, and the selected layer numbers are $\{2, 2, 6, 2\}$. We use the backbone weights pre-trained by the COCO dataset [45] and train a total of 12 epochs.

The classification loss $\mathcal{L}_{cls}$ and regression loss $\mathcal{L}_{reg}$ are computed using binary cross-entropy and smooth L_1 loss, respectively. Our template branch and search branch are mirrored to ensure consistency. We first feed the networks with the template image followed by the search image, and then vice versa. The total loss of this bi-directional process is used as the final complete loss.

When the VR-CNN outputs two bounding boxes $bbox_1$ and $bbox_2$ with the highest confidence scores $S_1 \in [0, 1]$ and $S_2 \in [0, 1]$ respectively, we update the template branch and the background distractor proposals D if the intersection over union (IoU) between $bbox_1$ and $bbox_2$ is greater than 0.8, and $S_1 + S_2 > 1.9$.

Evaluation Metrics: Three commonly used evaluation metrics, namely precision plot, success plot, and state accuracy, are utilized to assess the performance of all trackers through One-Pass Evaluation (OPE). Specifically, the precision plot calculates the percentage of frames in which the estimated target location falls within a given distance threshold from the ground truth. We use a threshold of 20 pixels to rank the trackers. The success plot measures the fraction of successful frames where the Intersection over Union (IoU) between the predicted bounding box and the ground truth is greater than a threshold ranging from 0 to 1. The Area Under Curve (AUC) of the success plot is used to rank the trackers. The state accuracy, as defined in Anti-UAV [2], reflects the average overlap ratio between the predicted bounding boxes and the ground truth across all sequences. This metric not only requires the tracker to accurately predict the position and scale of the target, but also to determine the state of the target's

presence in the current frame, and make a judgment when the target disappears from the field of view.

B. Experiments on Anti-UAV410

Evaluated Trackers: For evaluation, a total of 25 publicly available trackers are selected, which include recent RGB trackers, i.e., SiamFC++ [46], PrDiMP [47], Super_DiMP [48], KYS [49], GlobalTrack [42], SiamCAR [39], SiamBAN [38], Siam R-CNN [12], ROAM [50], TrDiMP [51], TransT [52], STMTrack [53], HiFT [54], Stark [55], KeepTrack [56], RTS [57], Unicorn [58], AiATrack [59], OTrack [60], SLT-TransT [61], TCTrack [62], ToMP [63], SwinTrack [64], JointNLT [65], SeqTrack [66], ARTrack [67], DropTrack [68], GRM [69], ROMTrack [70], MixFormerV2 [71] and some TIR trackers, i.e., MLSSNet [72], CMD-DiMP [73], MMNet [74]. The default parameters provided by the authors of these trackers are used in the evaluation experiment. Note that to implement the JointNLT tracker, we assign a natural language reference “uav flying in the infrared scene” to each video sequence of Anti-UAV410 dataset.

Not Training with Anti-UAV410: Table III displays the raw tracking performance results without training using the Anti-UAV410 training set, ranked by state accuracy. From this table, we can reach the following conclusions:

- i) Some trackers that utilize deeper networks for feature extraction, such as Stark-ST101, SwinTrack-Base and SeqTrack-B384, do not significantly outperform trackers that use more shallow features, such as Stark-ST50, SwinTrack-Tiny and SeqTrack-B256. This is attributed to the small size of the target and the limited number of pixels it occupies in the infrared image, which makes it challenging for deeper networks to effectively leverage their semantic representation capabilities.
- ii) It is noteworthy that trackers employing global detection methods, such as Siam R-CNN and GlobalTrack, perform exceptionally well. We attribute this to their ability to handle camera motion and re-detect targets that are lost from the screen, thereby improving overall tracking performance.
- iii) Trackers that have achieved leading positions in traditional tracking datasets, such as SwinTrack and MixFormerV2, do not necessarily excel in anti-drone tracking tasks in TIR videos. Because the challenges of anti-drone tracking are different from those of traditional tracking, the requirements for trackers will be different as well.
- iv) Trackers designed for TIR video sequences, such as MMNet, MLSSNet and CMD-DiMP, do not perform well in anti-drone tracking missions. This is because these trackers only emphasize compensating for the loss of color information in TIR videos, but do not enhance the ability to capture small targets and suppress background thermal noise interference.

Training with Anti-UAV410: To validate the effectiveness of our proposed training set in improving the performance of trackers for TIR UAV target tracking, we conducted comparison experiments on representative deep trackers with training using

TABLE III
RAW STATE ACCURACY (%) OF STATE-OF-THE-ART TRACKERS ON ANTI-UAV410 TEST AND VALIDATION SETS

Methods	Source	Test	Validation
RGB Trackers			
SiamFC++ [46]	AAAI20	41.03	45.15
PrDiMP50 [47]	CVPR20	51.87	53.35
Super_DiMP [48]	-	53.12	55.87
KYS [49]	ECCV20	44.70	49.61
GlobalTrack [42]	AAAI20	59.34	63.39
SiamCAR [39]	CVPR20	41.73	48.18
SiamBAN [38]	CVPR20	36.69	38.36
Siam R-CNN [12]	CVPR20	60.54	64.81
ROAM [50]	CVPR20	42.41	47.77
TrDiMP [51]	CVPR21	50.20	55.47
TransT [52]	CVPR21	48.85	55.24
STMTrack [53]	CVPR21	38.98	41.24
HiFT [54]	ICCV21	33.84	41.74
Stark-ST50 [55]	ICCV21	54.07	61.43
Stark-ST101 [55]	ICCV21	54.15	59.57
KeepTrack [56]	ICCV21	55.97	58.34
RTS [57]	ECCV22	54.68	57.19
Unicorn-Large [58]	ECCV22	56.55	59.38
Unicorn-Tiny [58]	ECCV22	55.42	58.93
AiATrack [59]	ECCV22	53.91	55.24
OTrack-256 [60]	ECCV22	43.75	52.60
OTrack-384 [60]	ECCV22	53.53	59.20
SLT-TransT [61]	ECCV22	50.90	57.62
TCTrack [62]	CVPR22	35.37	35.05
ToMP50 [63]	CVPR22	52.95	55.56
ToMP101 [63]	CVPR22	54.52	60.14
SwinTrack-Tiny [64]	NIPS22	51.23	56.83
SwinTrack-Base [64]	NIPS22	52.41	55.76
JointNLT [65]	CVPR23	48.92	54.36
SeqTrack-B256 [66]	CVPR23	52.87	56.38
SeqTrack-B384 [66]	CVPR23	49.97	53.01
ARTrack [67]	CVPR23	48.50	52.46
DropTrack [68]	CVPR23	52.99	59.78
GRM [69]	CVPR23	42.25	50.84
ROMTrack [70]	ICCV23	48.13	55.08
MixFormerV2-B [71]	NIPS23	55.22	59.71
TIR Trackers			
MMNet [74]	AAAI20	27.95	35.26
MLSSNet [72]	TMM'20	26.51	34.97
CMD-DiMP [73]	ACM MM21	52.75	56.18

the Anti-UAV410 training set. All re-trained trackers retain default settings and default pre-training weights, the experimental results are shown in Table IV. Specifically, the numbers on the left side of the table represent the results obtained using the original models provided by the authors, while the numbers on the right side indicate the results after training with our proposed training set. With the inclusion of our Anti-UAV410 training set, the performance of the trackers has been improved to a certain extent on both the validation and test sets. Remarkably there are six trackers that have improved their performance by more than 10% on both sets with the help of the Anti-UAV410 dataset. This clearly demonstrates the value of our dataset in enhancing the performance of trackers for anti-UAV applications.

The proposed SiamDT method ranks the first place with scores of 68.19 and 71.65 on the test and validation set, respectively. SiamDT has a slight performance gain over the global two-stage tracking methods GlobalTrack and Siam R-CNN,

TABLE IV
RESULTS OF REPRESENTATIVE DEEP BASED TRACKERS TRAINED USING ANTI-UAV410 TRAIN SET

Methods	Original model		Trained model			
	Training Datasets	Test	Validation	Training Datasets	Test	Validation
PrDiMP50	LaSOT, GOT10k, TrackingNet, COCO	51.87	53.35	Anti-UAV410	54.69(+ 5.4%)	62.52(+17.2%)
Super_DiMP	LaSOT, GOT10k, TrackingNet, COCO	53.12	55.87	Anti-UAV410	59.68(+12.3%)	64.56(+15.6%)
KYS	LaSOT, GOT10k, TrackingNet	44.70	49.61	Anti-UAV410	44.90(+ 0.4%)	49.91(+ 0.6%)
GlobalTrack	LaSOT, GOT10k, COCO	59.34	63.39	Anti-UAV410	66.45(+12.0%)	70.63(+11.1%)
SiamBAN	ImageNet, YouTube-BB, LaSOT, GOT10k, COCO	36.69	38.36	Anti-UAV410	47.32(+29.0%)	46.55(+21.4%)
Siam R-CNN	ImageNet, YouTube-VOS, LaSOT, GOT-10k	60.54	64.81	Anti-UAV410	63.00(+ 4.1%)	68.25(+ 5.3%)
Stark-ST101	LaSOT, GOT-10k, TrackingNet, COCO	54.15	59.57	Anti-UAV410	57.15(+ 5.5%)	64.18(+ 7.7%)
KeepTrack	LaSOT, GOT10k, TrackingNet, COCO	55.97	58.34	Anti-UAV410	56.80(+ 1.5%)	59.36(+ 1.7%)
AiATrack	LaSOT, GOT-10k, TrackingNet, COCO	53.91	55.24	Anti-UAV410	59.56 (+10.5%)	66.04(+19.6%)
TCTrack	ImageNet, LaSOT, GOT-10k	35.37	35.05	Anti-UAV410	41.64(+17.7%)	41.08(+17.2%)
ToMP50	LaSOT, GOT-10k, TrackingNet, COCO	52.95	55.56	Anti-UAV410	55.09(+ 4.0%)	64.30(+15.7%)
ToMP101	LaSOT, GOT-10k, TrackingNet, COCO	54.52	60.14	Anti-UAV410	55.10(+ 1.1%)	63.78(+ 6.1%)
SwinTrack-Tiny	LaSOT, GOT-10k, TrackingNet, COCO	51.23	56.83	Anti-UAV410	53.15(+ 3.7%)	60.84(+ 7.1%)
SwinTrack-Base	LaSOT, GOT-10k, TrackingNet, COCO	52.41	55.76	Anti-UAV410	55.74(+ 6.4%)	63.57(+14.0%)
DropTrack	LaSOT, GOT-10k, TrackingNet, COCO	52.99	59.78	Anti-UAV410	60.15(+13.5%)	66.16(+10.7%)
MixFormerV2-B	LaSOT, GOT-10k, TrackingNet, COCO	55.22	59.71	Anti-UAV410	59.65(+ 8.0%)	64.32(+ 7.7%)
SiamDT	-	-	-	Anti-UAV410	68.19	71.65

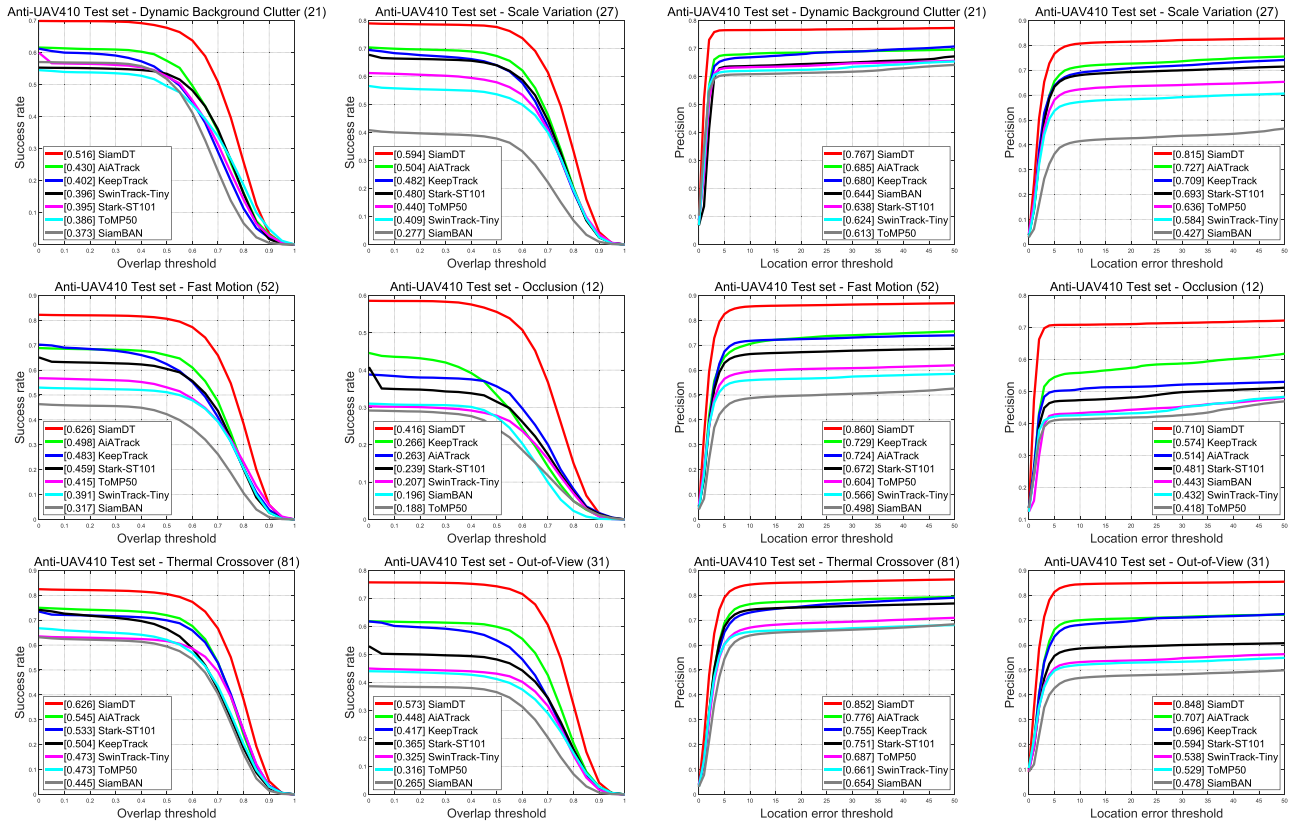


Fig. 4. Attribution evaluation on Anti-UAV410. Success plot (left) and precision plot (right) are presented. For optimal viewing experience, it is recommended to view the plots in color and zoomed in. Additional details regarding the definitions of attributes can be found in Table I.

and achieves a maximum accuracy improvement of nearly 30% compared to the TCTrack tracker. It is worth noting that SiamDT exceeds the re-trained Stark, TransT and ToMP algorithm on the test set by more than 10%, which are recent transformer-based tracking algorithms. What's more, SiamDT achieves over 10% performance gain compared to Swin Transformer-based tracker SwinTrack.

Per-Attribute Evaluation: As shown in Fig. 4, we provide a comprehensive evaluation of different trackers based on their

performance in different attributes using success and precision plots. We selected six representative trackers for comparison, all of which were trained using the Anti-UAV410 training set. Our SiamDT consistently performs the best across all six attribute evaluations.

Fast motion will blur the target, occlusion or out of view will pollute the target model, if no measures are taken to remove such interference, it will lead to irreversible errors. SiamDT performs well in these three challenges and achieves a large performance

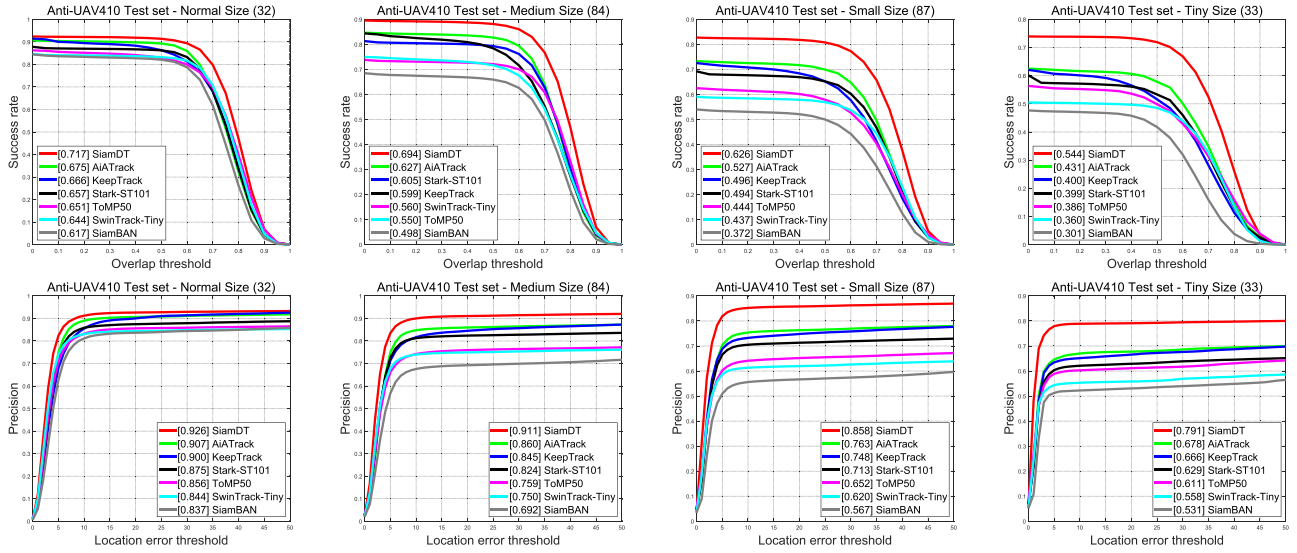


Fig. 5. Scale evaluation on Anti-UAV410. Success plot (top) and precision plot (bottom) are given. Better viewed in color and zoom.

gain over the compared trackers. The target size usually changes during anti-UAV tracking, from the figure we can see that the proposed SiamDT tracker can estimate the scale changes of the UAV more accurately.

Notably, in the dynamic background clutter and thermal crossover attribute, our SiamDT outperforms the advanced Swin Transformer-based tracker, SwinTrack, by a significant margin of more than 12% in success plot and 14% in precision plot. This demonstrates the effectiveness of our dual semantic features in mitigating the impact of thermal cluttered backgrounds, leading to improved tracking performance in challenging scenarios. It showcases the potential of our approach in tackling real-world tracking scenarios with complex backgrounds and dynamic environments.

Scale-based Evaluation: We conducted further analysis of the performance of different trackers in tracking Unmanned Aerial Vehicles (UAVs) of varying sizes. Videos were categorized into four groups, namely normal scale, medium scale, small scale, and tiny scale, based on the diagonal length of the target. The precision and success plots of the tracking methods for each scale category are depicted in Fig. 5. Notably, SiamDT achieved the top ranking in all size attributes, exhibiting a significant margin of superiority over the second-ranked tracker, AiATrack.

It is worth mentioning that as the scale of the UAVs becomes smaller, the lead of SiamDT over other trackers becomes more pronounced. This underscores the superiority of our approach in handling long-range small object tracking scenarios, which are commonly encountered in anti-UAV missions in uncontrolled environments. In the tiny scale category, SiamDT outperforms the Swin Transformer-based tracker, SwinTrack, by 23.3% and 18.4% in terms of precision and success, respectively.

Qualitative Comparisons: We have selected a set of representative tracking scenarios to visually showcase the performance of our tracking approach. As depicted in Fig. 6, SiamDT demonstrates its superiority over other trackers in effectively handling challenging tracking situations that are commonly encountered

in real-world anti-UAV tasks. This underscores the practical applicability of our approach, which has significant potential in practical deployment.

C. Ablation Studies

To verify the effectiveness of different components of our SiamDT, we conducted comprehensive ablation experiments and performed detailed analysis. Importantly, we kept the hyperparameters during training consistent to ensure a fair comparison. All experiments were conducted on the Anti-UAV410 test set and validation set. We used state accuracy (SA) to evaluate trackers, and details about the SA metric can be found in [2].

We used the following notations in our experiments: GS for global search from the whole image, MS for matching semantics of the first frame, FS for foreground probability semantics of drones, and BDS for background distractors suppression module. Additionally, we created two baseline methods, SiamMS and SiamFS. SiamMS is similar to traditional Siamese trackers that only exploit the matching semantics between the target and the template, while SiamFS is similar to detection methods that ignore spatio-temporal information of the sequence.

Effect of Global Search strategy: SiamNGS (Non Global Search, NGS) represents the same algorithm as SiamDT, but adopts the same neighborhood search strategy as SiamRPN to replace SiamDT's global search. Without global search, SiamDT will face a staggering 12.5% loss in tracking performance, demonstrating the importance of global search on anti-drone tracking missions. Because UAV targets are usually very small, this results in a relatively small percentage of the neighborhood search area and the full image, and a fast-moving UAV can easily escape from such search area, rendering the tracker completely invalid.

Effect of Background Distractors Suppression Module: SiamBDS represents the method that introduces the BDS module to SiamMS. As shown in Table VI, with the help of

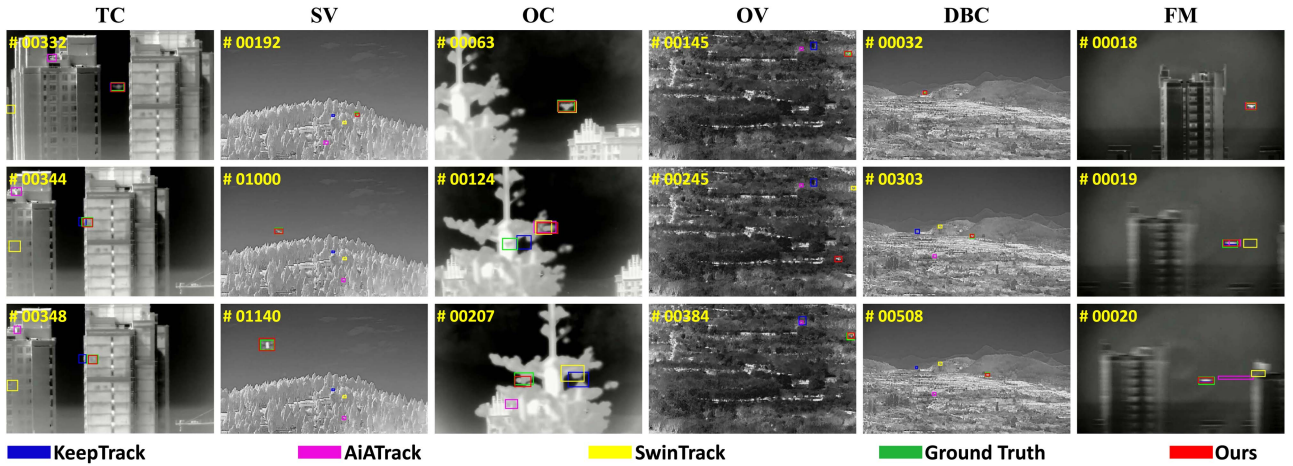


Fig. 6. Qualitative comparison. We selected challenging sequences from the Anti-UAV410 dataset and visualized the results of state-of-the-art trackers along with SiamDT. Comparisons are shown for 6 attributes: TC (Thermal Crossover), SV (Scale Variation), OC (Occlusion), OV (Out-of-View), DBC (Dynamic Background Clutter), FM (Fast Motion). Best viewed in color and zoom.

TABLE V
PERFORMANCE OF SIAMDT ON ANTI-UAV, LASOT, AND GOT-10 K DATASET

	Source	Anti-UAV SA	AUC	LaSOT P	P_{Norm}	GOT-10k AO	SR ₅₀	SR ₇₅
ATOM [75]	CVPR19	55.6	51.5	50.5	57.6	55.6	63.4	40.2
SiamRPN++ [41]	CVPR19	43.0	49.6	49.1	56.9	51.7	61.6	32.5
DiMP50 [76]	ICCV19	59.8	56.9	56.7	65.0	61.1	71.7	49.2
ROAM [50]	CVPR20	44.9	44.7	44.5	54.3	46.5	53.2	23.6
SiamCAR [39]	CVPR20	49.8	50.7	51.0	60.0	56.9	67.0	41.5
GlobalTrack [42]	AAAI20	71.3	52.1	52.9	59.9	-	-	-
SiamBAN [38]	CVPR20	51.0	51.4	52.1	59.8	-	-	-
SiamGAT [77]	CVPR21	-	53.9	53.0	63.3	62.7	74.3	48.8
TransT [52]	CVPR21	52.1	64.9	69.0	73.8	67.1	76.8	60.9
TrDiMP [51]	CVPR21	54.8	63.9	61.4	-	67.1	77.7	58.3
KeepTrack [56]	ICCV21	61.1	67.1	70.2	77.2	-	-	-
AiATrack [59]	ECCV22	58.9	69.0	73.8	79.4	69.6	80.0	63.2
SwinTrack-Tiny [64]	NIPS22	56.0	67.2	70.8	-	71.3	81.9	64.5
SwinTrack-Base [64]	NIPS22	57.2	71.3	76.5	-	72.4	80.5	67.8
SiamDT	Ours	72.2	57.7	60.1	68.3	64.2	72.2	57.0

TABLE VI
ABLATION STUDY OF DIFFERENT MODULES IN THE SIAMDT ON THE ANTI-UAV410 TEST SET

	GS	Increments MS	FS	BDS	Anti-UAV410 SA(%)
SiamNGS		✓	✓	✓	55.7
SiamMS	✓	✓			63.3
SiamFS	✓		✓		62.9
SiamBDS	✓	✓		✓	64.9
SiamDS	✓	✓	✓		64.2
SiamDT	✓	✓	✓	✓	68.2

GS: global search, ms:matching semantic, FS: foreground probability semantic, BDS: background distractors suppression.

the BDS module, SiamBDS obtains a gain of 1.6% in terms of accuracy on the Anti-UAV410 test set. This demonstrates that our BDS module effectively suppresses background distractors, enhancing the robustness of anti-UAV tracking.

Effect of Dual Semantic Module: We conduct experiments to verify the improvement of tracking performance with dual semantic information. SiamDS denotes the algorithm that

TABLE VII
EFFECT OF DIFFERENT BACKBONES IN SIAMDT

Backbones	test	val	Backbones	test	val
DarkNetV3	58.61	65.73	ResNeXt101	66.83	70.87
ResNet50	66.93	70.83	Swin-Tiny	68.19	71.65
ResNet101	66.74	70.60	Swin-Small	67.70	71.38
ResNeXt50	67.40	71.03	Swin-Base	67.40	71.26

combines SiamMS and SiamFS without the BDS module. From the comparison of SiamDS and SiamMS, SiamFS, with the help of a more powerful feature learning mechanism, SiamDS outperforms SiamMS (0.9%) and SiamFS (1.3%) on the Anti-UAV410 test set. This fully proves the improvement of tracking performance with dual semantic information, which can handle the challenge of tiny targets with less semantic information and boost the development of TIR tracking.

Effect of Different Backbones: We further investigated the effect of different backbone networks on the performance of TIR tracking. Table VII shows the performance of SiamDT using different backbones. We observe that SiamDT benefits from the

TABLE VIII
EFFECT OF DIFFERENT LOSS FUNCTIONS IN SIAMDT

Loss function	test	val	Loss function	test	val
L1	68.20	71.27	GIOU	68.11	71.49
Smooth L1	68.19	71.65	DIOU	68.04	71.33

recent popular Swin Transformer backbone. Additionally, we can see from the table that a deeper network does not necessarily lead to better tracking results. This is because small targets occupy fewer pixels, and deeper features have lower resolution, which can result in inaccurate tracking results.

Effect of Different Loss Functions: We further compare the effect of different loss functions on SiamDT, such as L1, Smooth L1, GIOU and DIOU. The results of the comparison are shown in Table VIII, where we can see that the different loss functions have little effect on the performance of the tracker. Therefore SiamDT does not impose a strict restriction on the use of loss functions.

D. Experiments on Other Datasets.

We further conducted experiments on three popular large-scale datasets, namely Anti-UAV [2], LaSOT [15], and GOT-10k [16], to validate the generality of our tracker. We trained SiamDT and other trackers listed in Table V on the Anti-UAV, LaSOT, and GOT-10 k training sets separately, and used each trained model for evaluation on its corresponding dataset. We used state accuracy (SA) to evaluate trackers on the Anti-UAV test set, Success (AUC), Precision (P), and Normalized Precision (P_{Norm}) to evaluate trackers on LaSOT, and Average Overlap (AO), Success Rate (SR_{50}), and Success Rate (SR_{75}) to evaluate trackers on GOT-10 k. More detailed definitions of these metrics can be found in Anti-UAV [2], LaSOT [15], and GOT-10k [16], respectively.

As presented in Table V, with the help of dual-semantic feature extraction and background distractors suppression mechanism, our SiamDT ranks the seventh place in accuracy when coping with multi-category target tracking on On the GOT10 k and LaSOT datasets. This demonstrates the generality and effectiveness of our proposed method in handling diverse tracking scenarios and target categories. On the Anti-UAV dataset, SiamDT ranks the first place and achieves an accuracy gain of more than 10% over most compared algorithms. Since our dual-semantic RPN and versatile RCNN are both designed for tracking UAV targets, our tracker is effective for single/few-class tracking tasks, while its application potential in tracking generic targets will not be remarkable.

VI. CONCLUSION

In this paper, we have focused on the problem of anti-UAV tracking. We first proposed a new large-scale TIR dataset called Anti-UAV410, which contains a variety of actual UAV videos, to provide a benchmark for evaluating anti-UAV tracking algorithms. Furthermore, we proposed a Siamese tracking method called SiamDT that can effectively handle the challenges of tiny targets and dynamic background clutter in anti-UAV tracking

scenarios. Extensive experiments conducted on Anti-UAV410 and other popular datasets demonstrated the effectiveness of our proposed method in achieving state-of-the-art performance in anti-UAV tracking.

In our future work, we plan to further enhance the dataset by adding more information about the camera, such as azimuth angle and pitch angle, to facilitate the establishment of motion models and improve the design of anti-UAV systems. This additional information can potentially enable better understanding of the UAV's motion patterns and lead to more robust tracking algorithms. We believe that our work will contribute to the development of the anti-UAV industry and pave the way for further advancements in this field.

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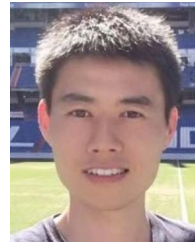
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